Perturbation Theory To All Orders

The Process

These notes explore the process of finding scattering amplitudes to an arbitrarily high order in g for φ^3.

The steps:

① Get $\Pi(k^2)$ and $V_3(k_1, k_2, k_3)$: we have to constantly adjust the counter-terms order by order in g to maintain $\Pi(-m^2)=0$, $\Pi'(-m^2)=0$ and $\vec{V}_3(0,0,0)=g$

② Construct the n-point vertex functions V_n where $4 \leq n \leq E$: we draw all 1PI diagrams omitting the ones with any loop corrections previously done. we then replace all 3 point vertices w. V_3 and propagators with the exact propagator. This is called skeletal expansion.

It is equivalent to computing V_n by summing all possible 1PI diagrams.

③ Get the necessary Tree-level diagrams: Sub in the loop corrected vertices and propagators and voila you have accounted for a LOT (all) possible diagrams.

④ Sum these diagrams to get your scattering amplitudes :)

Interestingly, we can think of all our altered tree diagrams as the normal Feynman Diagrams of a quantum action.